

Assignment 9

1. $f_B(x) \Rightarrow f_B(0) = 0$ & $f_B(1) = 1$

$$|0\rangle|0\oplus 0\rangle = |0\rangle|0\rangle$$

$$|0\rangle|1\oplus 0\rangle = |0\rangle|1\rangle$$

$$|1\rangle|0\oplus 1\rangle = |1\rangle|1\rangle$$

$$|1\rangle|1\oplus 1\rangle = |1\rangle|0\rangle$$

$f_C(x) \Rightarrow f_C(0) = 1$ & $f_C(1) = 0$

$$|0\rangle|0\oplus 1\rangle = |0\rangle|1\rangle$$

$$|0\rangle|1\oplus 1\rangle = |0\rangle|0\rangle$$

$$|1\rangle|0\oplus 0\rangle = |1\rangle|0\rangle$$

$$|1\rangle|1\oplus 0\rangle = |1\rangle|1\rangle$$

$f_D(x) \Rightarrow f_D(0) = 1$ & $f_D(1) = 1$

$$|0\rangle|0\oplus 1\rangle = |0\rangle|1\rangle$$

$$|0\rangle|1\oplus 1\rangle = |0\rangle|0\rangle$$

$$|1\rangle|0\oplus 1\rangle = |1\rangle|1\rangle$$

$$|1\rangle|1\oplus 1\rangle = |1\rangle|0\rangle$$

2. $H^{\otimes 2} |00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$
 $= \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

L.H.S = $\frac{1}{2} (H.H) (I \otimes X) |0\rangle|0\rangle$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

R.H.S = $(H.H) |0\rangle|1\rangle$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

L.H.S = R.H.S.

$$|+\rangle|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

So, $\frac{1}{2} (H.H) (I \otimes X) |0\rangle|0\rangle = (H.H) |0\rangle|1\rangle = |+\rangle|-\rangle$

3. $n=2$ & $|01\rangle_2 = |1\rangle_0$

$$H^{\otimes 2} |1\rangle = \frac{1}{2^{1/2}} \left((-1)^{0.1} |0\rangle + (-1)^{1.1} |1\rangle + (-1)^{2.1} |2\rangle + (-1)^{3.1} |3\rangle \right) \quad \text{--- (1)}$$

$$0.1 = (0 \text{ AND } 0) \oplus (0 \text{ AND } 1) = 0 \oplus 0 = 0$$

$$1.1 = (0 \text{ AND } 0) \oplus (1 \text{ AND } 1) = 0 \oplus 1 = 1$$

$$2.1 = (1 \text{ AND } 0) \oplus (0 \text{ AND } 1) = 0 \oplus 0 = 0$$

$$3.1 = (1 \text{ AND } 0) \oplus (1 \text{ AND } 1) = 0 \oplus 1 = 1$$

$$H^{\otimes 2} |1\rangle_0 = H^{\otimes 2} |01\rangle = \frac{1}{2} (|0\rangle - |1\rangle + |2\rangle - |3\rangle) \quad \text{--- (2)}$$

$$|+\rangle|-\rangle = \frac{1}{2} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$= \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \quad \text{--- (3)}$$

So, (2) = (3)

4. For balanced function $f(x)$ gives equal number of 0 & 1

ex. $f_0(0) = 0, f_0(1) = 1$

$$U_f |0\rangle |0\rangle = |0\rangle |0 \oplus f(0)\rangle = |0\rangle |0 \oplus 0\rangle = |00\rangle$$

$$U_f |0\rangle |1\rangle = |0\rangle |1 \oplus f(0)\rangle = |0\rangle |1 \oplus 0\rangle = |01\rangle$$

$$U_f |1\rangle |0\rangle = |1\rangle |0 \oplus f(1)\rangle = |1\rangle |0 \oplus 1\rangle = |11\rangle$$

$$U_f |1\rangle |1\rangle = |1\rangle |1 \oplus f(1)\rangle = |1\rangle |1 \oplus 1\rangle = |10\rangle$$

5. Given $x = 5, f(5) = 3, n = 3, m = 3$

y assumed as $|0\rangle_m$

$$U_f |x\rangle_n |y\rangle_m = |x\rangle_n |y \oplus f(x)\rangle_m$$

$$\begin{aligned} U_f |5\rangle_3 |0\rangle_3 &= |5\rangle_3 |0 \oplus f(5)\rangle_3 \\ &= |101\rangle_3 |0 \oplus 1\rangle_3 = |101\rangle_3 |100 \oplus 1\rangle_3 \\ &= |101\rangle_3 |011\rangle_3 \\ &= |101011\rangle \end{aligned}$$

6. Two basis vector $|a\rangle$ & $|b\rangle$ and inner product is $\langle a|b\rangle$

1) $|a\rangle$ transformed by XOR quantum oracle

We can assume y as 0

$$|a\rangle |0 \oplus f(a)\rangle = |a\rangle |f(a)\rangle$$

2) $|b\rangle \rightarrow |b\rangle |f(b)\rangle$

Inner product $\langle a|b\rangle$ is $\langle f(a)|f(b)\rangle \langle a|b\rangle$

If $a = b$ then we know inner product of same vector is

$$\langle f(a)|f(b)\rangle \langle a|b\rangle$$

$$\langle f(a)|f(a)\rangle \langle a|a\rangle$$

$$= 1 \cdot 1 = 1$$

If $a \neq b, \langle a|b\rangle = 0$

Basis vector Inner product is not changed. It is still unitary.

7.
$$V_{\text{swap}} = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} |j\rangle \langle i| |i\rangle \langle j|$$

$$\langle 00|00\rangle, N=4$$

$$\begin{aligned} &\langle 00|V_{\text{swap}}|00\rangle \langle 00| + \langle 01|V_{\text{swap}}|00\rangle \langle 01| + \langle 10|V_{\text{swap}}|00\rangle \langle 10| \\ &+ \langle 11|V_{\text{swap}}|00\rangle \langle 11| \end{aligned}$$

$$\langle 00|10\rangle$$

$$\begin{aligned} &\langle 00|V_{\text{swap}}|01\rangle \langle 00| + \langle 01|V_{\text{swap}}|01\rangle \langle 01| + \langle 10|V_{\text{swap}}|01\rangle \langle 10| \\ &+ \langle 11|V_{\text{swap}}|01\rangle \langle 11| \end{aligned}$$

$$\langle 00|V_{\text{swap}}|10\rangle \langle 00| + \langle 01|V_{\text{swap}}|10\rangle \langle 01| + \langle 10|V_{\text{swap}}|10\rangle \langle 10|$$

$$+ \langle 11|V_{\text{swap}}|10\rangle \langle 11|$$

$$\langle 00|V_{\text{swap}}|11\rangle \langle 00| + \langle 01|V_{\text{swap}}|11\rangle \langle 01| + \langle 10|V_{\text{swap}}|11\rangle \langle 10|$$

$$+ \langle 11|V_{\text{swap}}|11\rangle \langle 11|$$

$$= \langle 00|00\rangle |00\rangle \langle 00| + \langle 10|10\rangle |10\rangle \langle 10| + \langle 01|01\rangle |01\rangle \langle 01| + \langle 11|11\rangle |11\rangle \langle 11|$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = V_{\text{swap}}$$

8. $f_c(0) = 1, f_c(1) = 0$

$$U_f |0\rangle|0\rangle = |0\rangle|0 \oplus f(0)\rangle = |0\rangle|0 \oplus 1\rangle = |0\rangle|1\rangle$$

$$U_f |0\rangle|1\rangle = |0\rangle|0\rangle$$

$$U_f |1\rangle|0\rangle = |1\rangle|0\rangle$$

$$U_f |1\rangle|1\rangle = |1\rangle|1\rangle$$

$$\begin{aligned} & \langle 00 | U_f | 00 \rangle | 00 \rangle \langle 00 | + \langle 01 | U_f | 00 \rangle | 01 \rangle \langle 00 | + \langle 10 | U_f | 00 \rangle | 10 \rangle \langle 00 | \\ & + \langle 11 | U_f | 00 \rangle | 11 \rangle \langle 00 | \end{aligned}$$

$$\begin{aligned} & \langle 01 | U_f | 00 \rangle | 01 \rangle \langle 00 | + \langle 00 | U_f | 01 \rangle | 00 \rangle \langle 01 | + \langle 10 | U_f | 10 \rangle | 10 \rangle \langle 10 | \\ & + \langle 11 | U_f | 11 \rangle | 11 \rangle \langle 11 | \end{aligned}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} |00\rangle \rightarrow |01\rangle \\ |01\rangle \rightarrow |00\rangle \\ |10\rangle \rightarrow |10\rangle \\ |11\rangle \rightarrow |11\rangle \end{array} \quad \begin{array}{l} q_0 \oplus \oplus \oplus \oplus \\ q_1 \oplus \oplus \oplus \oplus \end{array}$$

$$P_D = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} |00\rangle \rightarrow |01\rangle \\ |01\rangle \rightarrow |00\rangle \\ |10\rangle \rightarrow |11\rangle \\ |11\rangle \rightarrow |10\rangle \end{array} \quad \begin{array}{l} q_0 \oplus \oplus \oplus \oplus \\ q_1 \oplus \oplus \oplus \oplus \end{array}$$